

Arithmetic Sequences as Linear Functions

Algebra 1

Writing recursive rules from listed sequences and from applications, using a recursive rule to reach a later term, and rewriting a recursive rule as an explicit linear function

Directions. A recursive formula needs *two* lines: the starting value a_1 and the step rule $a_n = a_{n-1} + d$ for $n \geq 2$, where d is the common difference. **Write both lines for every problem** — with units where the problem has them — and then answer the question that follows. The answer line is for the final numerical answer.

1. A growing tile pattern uses this many tiles in each stage:

$$4, \quad 11, \quad 18, \quad 25, \quad \dots$$

Write the recursive formula for a_n , the number of tiles in stage n , then find the number of tiles in stage 6.

Answer: _____

2. A drone descends in equal steps. Its height in meters at each reading is

$$63, \quad 58, \quad 53, \quad \dots$$

Write the recursive formula for a_n , the height at reading n , then find the height at reading 9. (Watch the sign of d .)

Answer: _____

3. A tank already holds 240 liters of water at the end of hour 1, and a pump adds 85 liters during each hour after that.

Write the recursive formula for a_n , the number of liters in the tank at the end of hour n . Then find how much water the tank holds at the end of hour 5.

Answer: _____ L

4. A sequence is defined by

$$a_1 = 12, \quad a_n = a_{n-1} + 9 \quad (n \geq 2).$$

Say in one sentence what this rule tells you to do, then find a_{10} .

Answer: _____

5. A sequence is defined by

$$a_1 = 7, \quad a_n = a_{n-1} + 4 \quad (n \geq 2).$$

Because the same amount is added each step, this sequence is a linear function of n . Rewrite it as an explicit rule $a_n = mn + b$, state what m and b mean in terms of the recursive rule, and use the explicit rule to find a_{20} .

Answer: _____

6. A sequence is defined by

$$a_1 = 5, \quad a_n = a_{n-1} + 6 \quad (n \geq 2).$$

One term of this sequence equals 89. Which term is it? Give the term number n .

Answer: _____

7. One chair is 84 centimeters tall. Each additional chair stacked on top of the pile adds 9 centimeters to the total height.

Write the recursive formula for a_n , the height in centimeters of a pile of n chairs. Then find the height of a pile of 12 chairs.

Answer: _____ cm

8. An arithmetic sequence is recorded in this table, but the first two terms were lost.

n	1	2	3	4	5	6	7
a_n	?	?	20				44

Find the common difference from the two known terms, write the recursive formula, and give the missing value of a_1 .

Answer: _____

9. A student's savings account holds 45 dollars at the end of week 1, and the student deposits 18 dollars at the end of every week after that.

Write the recursive formula for a_n , the balance in dollars at the end of week n . In which week does the balance first reach 405 dollars? Give the week number.

Answer: _____

10. Synthesis. A theater is built so that row 1 has 22 seats and each row after it has 3 more seats than the row in front of it.

(i) Write the recursive formula for a_n , the number of seats in row n . (ii) Explain in one sentence why a recursive rule alone is awkward for this next question. (iii) Find the total number of seats in the first 15 rows.

Answer: _____